Leigh Joiner - On The Flipside

Concept:

The primary mechanics of my game are the “flip” and backwards controls. While gameplay is normal on one side, at the press of a button, the whole world becomes a mirrored version of itself, flipping not only the platforms and adding some items/enemies, but also the user interface. My concept is to have the player work on two “planes” in order to solve puzzles and progress.

Target Audience:

The target audience is children and teens who enjoy puzzle games such as Pico Park, with a challenge that can be comparable to Mario Maker.

Construction Guide:

Main mechanic:

The main mechanic is the “flip” and backwards controls. (Detailed further in “The ‘Flip’” and “Player Character”.)

The Normal world:

The plane the player starts off on. It is visually brighter than the Other world. Fly, Fish, and Slime enemies may appear in levels. Enemies here may hurt the player and reset the level if hearts reach 0. When in this plane, left (<) and right (>) are normal.

The “Flip”:

At the press of a button (Y on controller, Tab on keyboard), the world physically flips around. To achieve this, the sides of the screen will come into the center, and the Other world will take its place, starting at the center of the screen and expanding to the sides. This simple illusion makes it seem as though the screen has physically flipped.

The Other world:

This plane is a complete mirror of the Normal world. Exceptions to this may include enemy types, buttons/levers in different locations, objects that do not appear on the Normal side, etc. These change depending on level and puzzle solutions.

Fly, Fish, and Snail enemies may appear in levels. Enemies here may hurt the player and reset the level if hearts reach 0. When in this plane, left (<) and right (>) are reversed; left (>), right (<). The jump button (A or Space), and the Flip button (Y) remain the same.

Camera:

The camera should fit the entire level and remain static. Size of the camera should expand or shrink to accommodate the level size.

Player Character(s):

Move (Normal):

Left: <
Right: >
Jump: A or Space
Flip worlds: Y or Tab

Move (Other):
Left: >
Right: <
Jump: A or Space
Flip worlds: Y or Tab

The character on either side should move independently from each other. If Normal walks to the left 2 paces, Other should still be in the location it was in previously (ie. not 2 paces to the right (directions mirrored)).

Objects:

Door:

The door's Open condition will change each level. The level begins with the door shown Closed. If the level's conditions are met, the door will appear Open. Going near/moving in front of the open door will cause the Player to fade into it (to appear to be going through it).

The player character on the Normal and Other sides must both go through the door on their sides. One side going through is not enough to activate the win condition.

If one door is not Open, the other side's player character does not go through the door. Player is only allowed through the doors when both are unlocked and Open.

Buttons:

Buttons appear Unpressed at the start of level, unless otherwise altered for the level. If the effect the button changes must be active at the beginning of a level (ie. buttons on the Normal and Other sides requiring one be Pressed and the other Unpressed for the purpose of a puzzle), that button begins in the level as Pressed.

Crates can be used to press a button as long as it remains there. Players standing on the button will also press it. Some levels may require an object or Player remain on the button to keep it Pressed, while some only require an object or Player to interact with it once.

Buttons can activate any number of effects as needed, such as: Unlocking doors, revealing platforms, dropping items, etc.

Levers:

Levers may appear Straight, Left, or Right, depending on the requirements of the level. Straight (when the lever points directly up) may be used when it needs to be pulled in one direction or the other to change platform direction, control physics object(s), etc. Left or Right may be used

if the lever requires it to be activated only once (at which case it switches positions (Left to Right, Right to Left)).

Remember: The Other world is a direct mirror of the Normal world. If a lever in the Normal world is to the Left, the lever in the Other world must be to the Right. However, if the Normal Player interacts with the lever (Left to Right), the Other world lever may not be changed, depending on the requirements of the level.

Keys:

Keys can be collected and used to unlock doors in some levels. To collect a key, the Player must collide with it. It hovers over the Player until they collide with a door. The key then hovers above the door and it Opens.

Crates:

Crates are physics objects that can be pushed by the Player as needed to solve puzzles. A crate may appear on one side (Normal or Other), and not the other. This varies by level. If a crate is pushed on one side, it may or may not move on the other, also varied by level.

Crates can be pushed on top of buttons. When there, the button will remain pressed until it is pushed off.

Water/Lava:

Water and Lava have largely the same mechanics and so will be discussed as one. The Player cannot swim. If the Player jumps or falls into the water/lava and does not get out within a couple seconds (~2 seconds), the Player loses health. For every second they remain in the water/lava, another tick of health is lost. If they jump out and back into the water/lava, the 2 second timer resets. If the Player's health reaches 0, the level resets.

Enemies may appear in the water/lava.

Enemies:

All collisions with enemies cause damage to Player.

Fish:

Fish appear in water or lava, swimming back and forth within a designated area.

Slime/Snail:

Slime and Snail enemies walk within a designated area in levels. Slime enemies only appear in the Normal world. Snail enemies only appear in the Other world.

Fly:

Fly enemies hover in the air within a designated area.

User Interface:

Controls:

Controls are displayed at the top of the screen. When the worlds are Flipped, the controls are also flipped, but any text remains readable, only switching sides of the screen.

The arrow buttons displayed (Left and Right), also show the flip in control mechanics when in the Other world by retaining their name label (“Left”, “Right”), but switching the directions they are associated with.

Hearts:

Hearts appear in the top Right (Normal) or Left (Other) next to the “Flip” button. In the Normal world, Damage is taken from the left-most heart. In the Other world, Damage is taken from the right-most container.

Lock/Unlock:

In the top Right (Normal) or Left (Other) next to the “Flip” button is a padlock symbol. This shows the Player if the Door on that side is Locked or Unlocked.

If the Player tries to exit the level through an Open door while one door on either side remains Closed, the padlock symbol will shake, to tell them they cannot leave without unlocking both doors.

Warning:

If a Player Character is in the path of an enemy when the worlds are Flipped, an exclamation point symbol appears in the top middle of the screen. The symbol pulses yellow. When the health of the Player Character reaches halfway, it pulses red.

Scope of Demo:

The demo will include 2-3 levels, guiding the player through a tutorial. They will be able to interact with the environment (enemies and objects), switch worlds, and solve puzzles to progress to other levels. Controls will be keyboard and controller compatible.

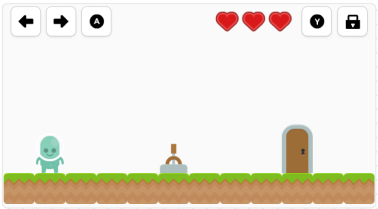
Resources:

Kenney game assets from https://opengameart.org/

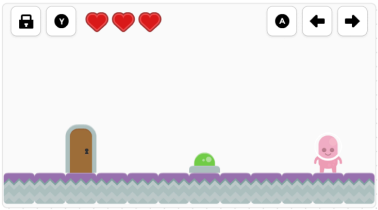
Level 1
Introductory level to get used to the controls and objective of the game

Controls are shown for controller, not mouse + keyboard

Normal world



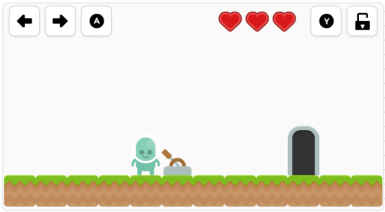
Other world



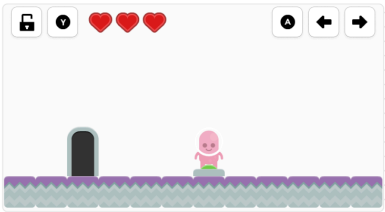
Solution

Button opens Normal door, lever opens Other door

Normal world

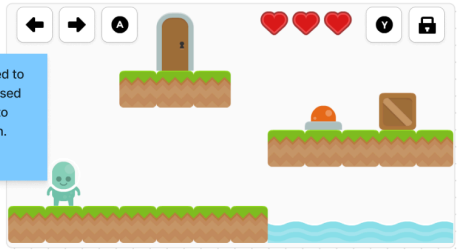


Other world

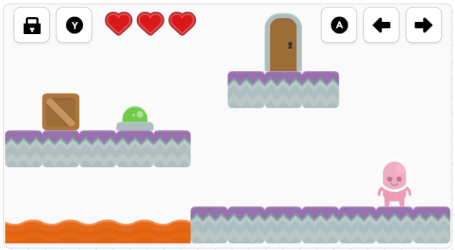


Buttons need to remain pressed for door(s) to remain open.

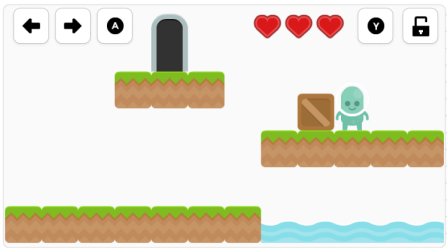
Normal world



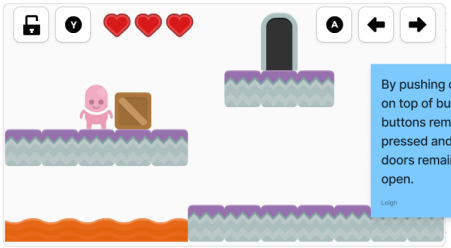
Other world



Normal world



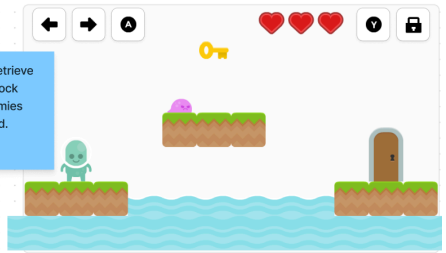
Other world



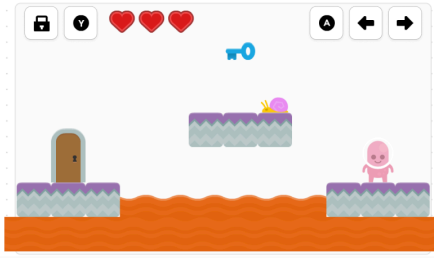
By pushing crates on top of buttons, buttons remain pressed and doors remain open.

Need to retrieve key to unlock door. Enemies introduced.

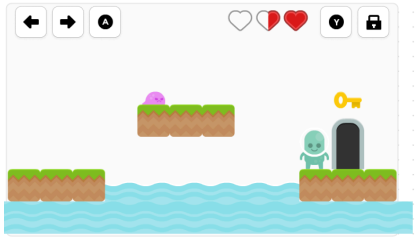
Normal world



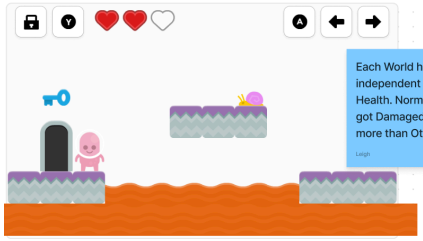
Other world



Normal world

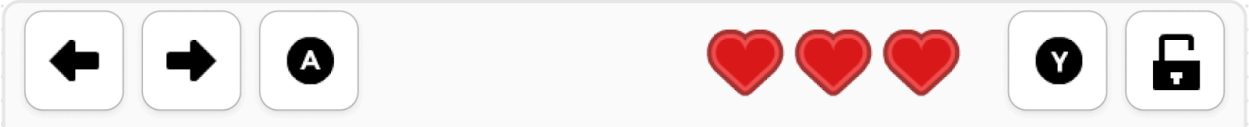


Other world

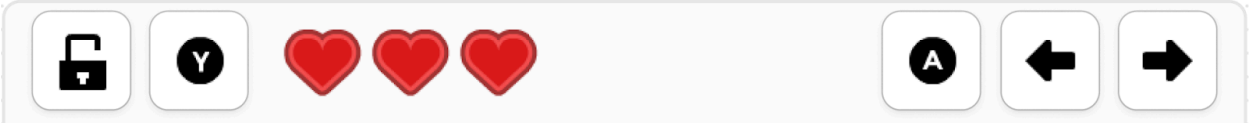
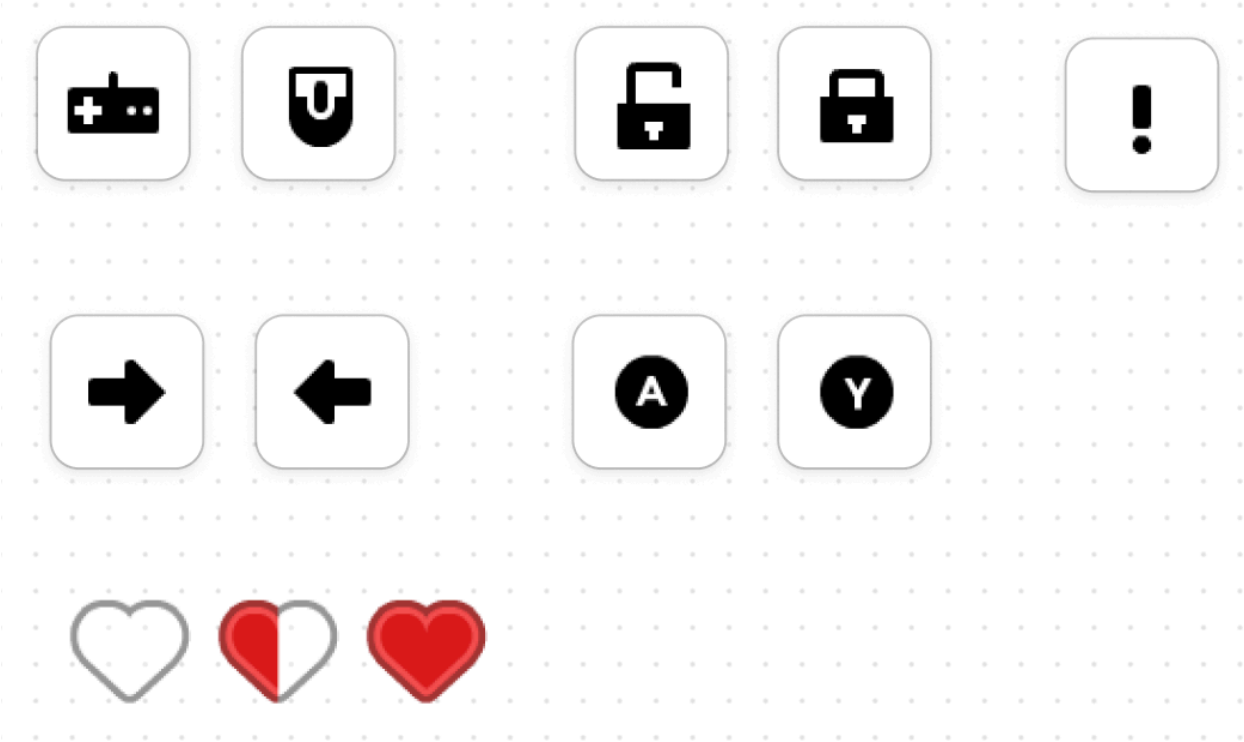


Each World has independent Health. Normal got Damaged more than Other.

UI of Normal World



UI of Other World



UI Symbols